

AIG130 – Lab 4

Objective

Explore and compare key features of AWS SageMaker, Google AutoML, and Azure ML Studio in solving real-world machine learning problems.

Group Formation

Students can work individually or form groups of up to 4 members. Collaboration and effective communication are encouraged for those working in groups.

Assignment

Choose a machine learning application. How would you use AWS SageMaker, Google AutoML, and Azure ML Studio to propose a solution? Reflect on the strengths and limitations of each platform. This is only an exploratory task and actual implementation is not required. The objective is to be exposed and be able to compare these cloud services.

Suggested Steps

1. Case Selection:
 - a. You can choose a case from a range of sectors such as healthcare, finance, retail, or technology.
2. Platform Exploration:
 - a. How can AWS SageMaker be used to implement the entire pipeline (as we discussed last week) of your proposed ML application?
 - b. Do the same with Google AutoML and Azure ML Studio
3. Comparative Analysis
 - a. Evaluate the ease of use, flexibility, and scalability of each platform.

Deliverables

Report: Submit a PDF report covering all three tasks. Include:

- An introduction and conclusion.
- Sections for each task with headings.
- Visual elements (charts, tables) where applicable.

Presentation: Prepare and deliver a presentation of your findings in class. The presentation is also required to be submitted in Blackboard. It can be as PDF, PPTX, Link, etc.

- Duration: Up to 15 minutes per group.
- Cover key points and findings from the report.
- Encourage questions and discussions.

Individual Reflection

Each student must submit a separate reflection document answering the following:

1. If you worked individually, why did you choose to go solo? What challenges did you encounter?
2. If you did it as a group, how were tasks distributed, and what was your specific task? What challenges did you encounter?
3. Why do you think ML Engineering is important for effective implementation of ML projects? There is no right or wrong answer. Explain and elaborate on the answer that you give.

Submission

Submissions are done through the provided links in Blackboard. There are three links:

1. Group report (one submission per group): 4%
2. Individual reflection: 2%
3. Presentation file/link: 2%

Assessment Criteria

- Quality and depth of research and analysis.
- Clarity and organization of the report.
- Effectiveness of the presentation.
- Thoughtfulness of the individual reflection.